

RESEARCH METHODS: QUALITATIVE

COMM50 Postgraduate Study & Research Methods

Qualitative Research (Salkind 2013)



*“...can be a powerful and appropriate non experimental way to explore an academic question rigorously, as **when additional context is needed to explain phenomena missed by quantitative research methods.** When properly performed, qualitative research projects add to the body of knowledge on their subjects and make the researcher **even more well informed.**”*

The Need for Qualitative



HOW DO YOU FEEL?



What is Qualitative Research?

Qualitative research is a *situated activity* that locates the observer in the world. It consists of a set of interpretive, material practices that *make the world visible*. These practices *transform the world*. They turn the world into a series of *representations*, including field notes, interviews, conversations, photographs, recordings and memos to the self.

...

This means that qualitative researchers study things in their *natural settings*, attempting to make sense of, or to *interpret*, phenomena in terms of the *meanings people bring* to them ... It is understood, however, that *each practice makes the world visible in a different way*.

(Denzin and Lincoln, 2005)

What is Qualitative Research?

- Concerned with *quality*, rather than *quantity* of research data
- Research activity often stems from questions
- Conducted considering principles of
 - ▣ Naturalism
 - ▣ Human understanding
 - ▣ Discovery / theory forming
- Typically inductive: general to the specific
- Subjective, often interactive, relationship between the researcher and participant(s)

Benefits of a Qualitative Approach



- 'Real world' scenarios
- Large amounts of descriptive, rich information
- Useful in describing and explaining complex objects and issues
- Removed objectivity
 - ▣ Impossible to be objective anyway?
- Does not attempt to simplify
- Allows for multiple perspectives (truths)

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Qualitative Research

- Qualitative research deals with investigating reasoning and things that cannot/are impossible to measure
- Answers questions of *Why?* and *How?*
- Common when dealing with humans and their opinions / behaviour
 - ▣ Interviews, focus groups, observation, etc.
- Experiments, more likely, studies are usually conducted in natural environments, not strictly controlled
- Time-consuming and usually smaller sample sizes than quantitative methods

Purposes of Qualitative Studies (Peshkin 1993)

- **Description.** They can reveal the multifaceted nature of certain situations, settings, processes, relationships, systems, or people.
- **Interpretation.** They enable a researcher to (a) gain new insights about a particular phenomenon, (b) develop new concepts or theoretical perspectives about the phenomenon, and/or (c) discover problems that exist within the phenomenon.
- **Verification.** They allow a researcher to test the validity of certain assumptions, claims, theories, or generalizations within real-world contexts.
- **Evaluation.** They provide a means through which a researcher can judge the effectiveness of particular policies, practices, or innovations.

Discussion!



*What aspects of your
computing discipline
computer might warrant a
qualitative approach?*

Qualitative Methods in Computing



Some possible investigations:

- *What effect has social networking had on workplace productivity?*
- *How can assisted-living technologies be used to improve the lives of children with disabilities?*
- *Why do computer games players exhibit aggressive tendencies?*
- *What is it like to be a network manager in the largest Internet search companies?*

Qualitative Research: Subjectivity

- Qualitative is an exploratory technique
- Subjectivity is present in the data collected from the subject AND interpretation of data by the researcher
 - ▣ Inherently, qualitative research is criticised as being biased
 - ▣ Researcher tries to find what they are looking for
 - ▣ Consider cultural expectations of research/participant
- A counter to this is to start with a broad question and form theory after the research
 - ▣ This is *grounded theory*

Qualitative Research: Subjectivity

- The Rorschach Inkblot Test



Qualitative Research: Subjectivity



- The Rorschach Inkblot Test
- As a projective test, it is deliberately subjective on the part of the participants
- A response to the previous inkblot may be “*bra*” [item of female clothing]
 - ▣ Male researchers are shown to interpret this as a “sex” classified response
 - ▣ Female researchers are shown to interpret this as a “clothing” classified response

Qualitative Methods



- Case Study
- Ethnography
- Phenomenology
- Grounded Theory
- Content Analysis

Qualitative Methods: Case Study

- Study of an event, individual or action over a fixed time window
- Useful in preliminary research and better defining a question
- Can be single or multiple (providing comparison)
- Focuses on a particular instance: results hard to generalise
- **Methods:** interviews; observations; document analysis; records; video; photographs; audio recordings; etc.
- **Analysis:** timelines; data categorisation; single instance examinations; identifying patterns; generalisation/synthesis

Qualitative Methods: Ethnography

- Study of a cognate group activity or experience
- Duration is lengthy (months or years)
- Captures the day-to-day
- Attempts to describe behaviour and reasons/beliefs
- Very hard to provide objective interpretations
- **Methods:** site observation / fieldwork – ideally a unfamiliar environment, accessed by a contact/gatekeeper; key informants; observation; field notes/diary; breaching experiments
- **Analysis:** description of events; narrative; timelines; pattern analysis; critical events;

Qualitative Methods: Phenomenological Study

- Attempt to describe a particular person or group's perspective or experience of a reality
 - ▣ *“What is it like to....”*
- Typically working individually with participants, several hours with each
- Researcher strives to be objective in data collection and analysis
- **Methods:** (unstructured) interviews (n=5 to 25)
- **Analysis:** interview transcribing; identification of themes and statements; grouping statements into meanings; identifying differences in perspectives; description formulating based on shared views

Qualitative Methods: Grounded Theory

- Formulation of theories and frameworks based upon data and information that already exists
- Useful where existing theories are shown to be invalid or where no theory exists
- Useful in studying processes and interactions
- Rigorous and objective – too much so?
- **Methods:** field-based; interviews; observations; document analysis; records; video; photographs; audio recordings; categorisation; saturation of categories; constant comparative method
- **Analysis:** open coding (characteristics of categories); axial coding (links between [sub]categories); selective coding (formation of a storyline or timeline); theory forming

Qualitative Methods: Content Analysis

- Detailed study of a information on a particular topic
- Often, the object(s) of study are communications (newspapers, books, magazines, TV programmes, Internet sites, and so on)
- Identifying patterns, themes, biases
- Requires fore planning and clear definition of boundaries
- Highly systematic and objective
- **Methods:** identification (entire body or sample); definition of characteristics being analysed; rating or seeking agreement on a characteristic's status
- **Analysis:** frequency analysis; tabulation; statistical analysis

TABLE 1 Distinguishing characteristics of different qualitative designs

<i>Design</i>	<i>Purpose</i>	<i>Focus</i>	<i>Methods of Data Collection</i>	<i>Methods of Data Analysis</i>
Case study	To understand one person or situation (or perhaps a very small number) in great depth	One case or a few cases within its/their natural setting	• Observations • Interviews • Appropriate written documents and/or audiovisual material	• Categorization and interpretation of data in terms of common themes • Synthesis into an overall portrait of the case(s)
Ethnography	To understand how behaviors reflect the culture of a group	A specific field site in which a group of people share a common culture	• Participant observation • Structured or unstructured interviews with "informants" • Artifact/document collection	• Identification of significant phenomena and underlying structures and beliefs • Organization of data into a logical whole (e.g., chronology, typical day)
Phenomenological study	To understand an experience from the participants' points of view	A particular phenomenon as it is typically lived and perceived by human beings	• In-depth, unstructured interviews • Purposeful sampling of 5–25 individuals	• Search for <i>meaning units</i> that reflect various aspects of the experience • Integration of the meaning units into a seemingly typical experience
Grounded theory study	To derive a theory from data collected in a natural setting	A process, including human actions and interactions and how they result from and influence one another	• Interviews • Any other relevant data sources	• Prescribed and systematic method of coding the data into categories and identifying interrelationships • Continual interweaving of data collection and data analysis • Construction of a theory from the categories and interrelationships
Content analysis	To identify the specific characteristics of a body of material	Any verbal, visual, or behavioral form of communication	• Identification and possible sampling of the specific material to be analyzed • Coding of the material in terms of predetermined and precisely defined characteristics	• Tabulation of the frequency of each characteristic • Descriptive or inferential statistical analyses as needed to answer the research question

Qualitative Data Collection



- Interview
- Observation
- Focus Group
- Questionnaire

Qualitative Data Collection: Interview

- Allows deep, detailed insight and directional, but time consuming
- Less natural environment
- Obedience / expectation an issue (Hawthorne effect)
- Structured
 - ▣ Constrained set of questions & responses
 - ▣ More precise, easier to analyse, may not be broad enough
- Semi-structured
 - ▣ Set questions but freedom in follow-up
 - ▣ Maintains consistency over subjects, more exploratory
- Unstructured
 - ▣ Highly naturalistic, conversational, but still controlled
 - ▣ Large data gathered, lots of analysis needed

Qualitative Data Collection: Interview

What makes a good (and bad) interview?



<http://youtu.be/9t-hYjAKww>

3m45s

Qualitative Data Collection: Observation

- Avoids intrusion/disturbance, supports naturalism
- Hard for researcher to detach from scenario
- Lots of information to deal with and process
- Able to work with very large groups
- Participant observation
 - ▣ The researcher is part of the 'group' and participates
 - ▣ Provides close observation but can become too close
- Non-participant observation
 - ▣ The researcher is detached (or hidden) from the 'group'
 - ▣ More naturalistic but ethical issues arise

Qualitative Data Collection



□ Focus Group

- ▣ As interview, but more participants
- ▣ Examination of group dynamic & individuals
- ▣ Danger of group being skewed by dominance
- ▣ Often produce “bland” results / compliance

□ Questionnaire

- ▣ Useful initial investigative tool
- ▣ Very large number of participants
- ▣ Not very natural – far from it
- ▣ Hard to ensure context is communicated

Discussion!



Can you give an example of a research scenario using a form of:

- 1. Interview?*
- 2. Observation?*
- 3. Focus group?*
- 4. Questionnaire?*

Qualitative: Subjects and Sampling



- Ethics are of paramount importance
 - ▣ Need for informed consent
 - ▣ Rights to privacy
- Sampling should be representative of a population
- But, in qualitative, a selective approach may be used to ensure contrasting views can be established

Qualitative: Subjects and Sampling



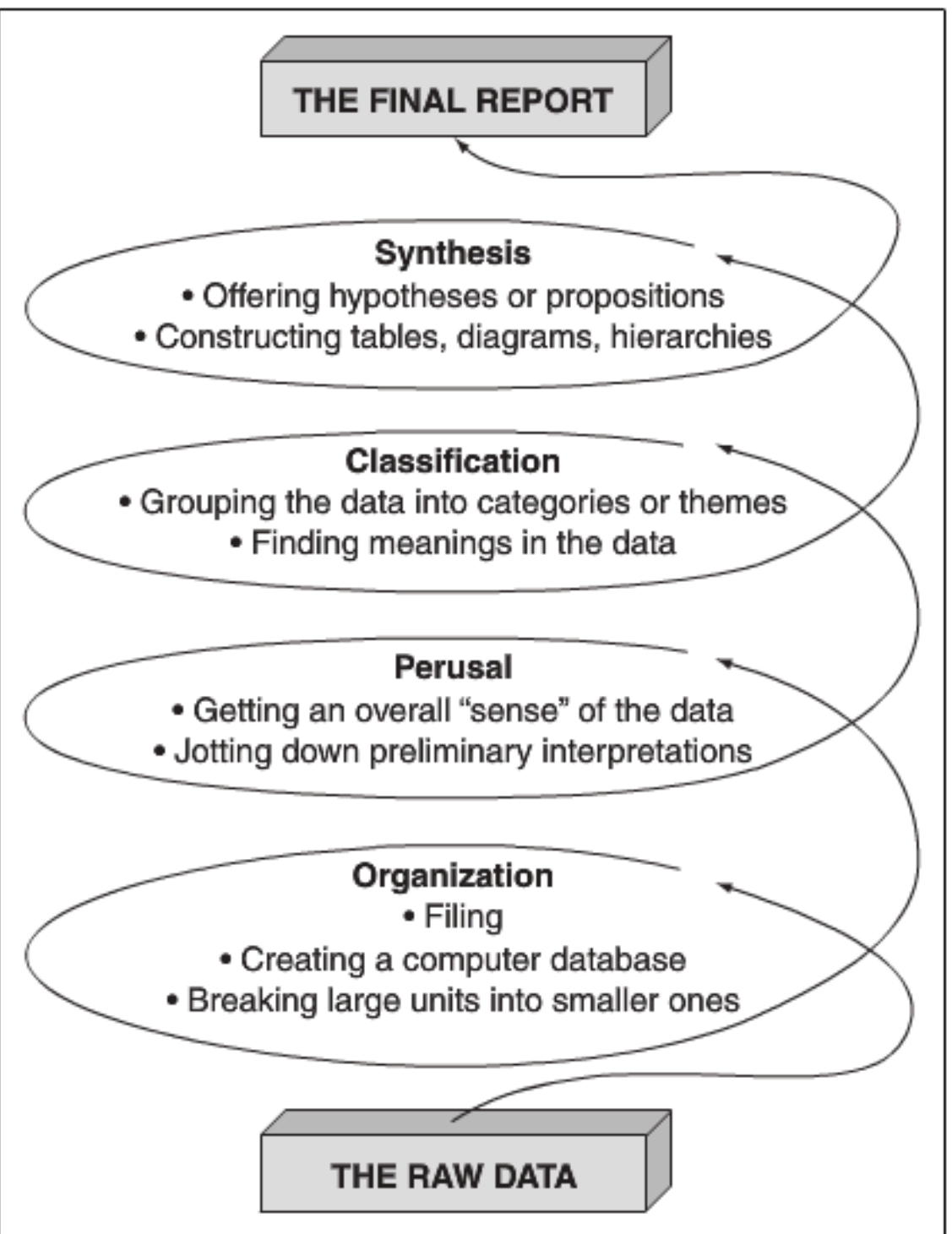
Leedy & Ormond (2013) provide sampling guidance:

- ▣ Be sure that the sample includes not only seemingly “typical” but also seemingly “non- typical” examples.
- ▣ When a power hierarchy exists - as it does in the workplace and in many clubs and communities - sample from various levels in the hierarchy. For example, in the work-place, interview both bosses and employees; in a club or community, interview not only highly active, influential members but also less involved individuals
- ▣ Actively look for cases that can potentially discredit emerging hypotheses and theories.

Towards Analysis of Qualitative Data

- Reading and re-reading of data
- Collating and correlating responses to theories
- Identification of themes:
 - ▣ Common; Conflicting; Minority
- Determine the features that join themes
- Formulate generalisations / abstractions of the data set and research studies
- Provide theories, evidenced by the research
- Can be assisted by computer analysis
 - ▣ Word searches, frequency counts, etc.

Cresswell's (2007) Spiral of Qualitative Analysis and Reporting



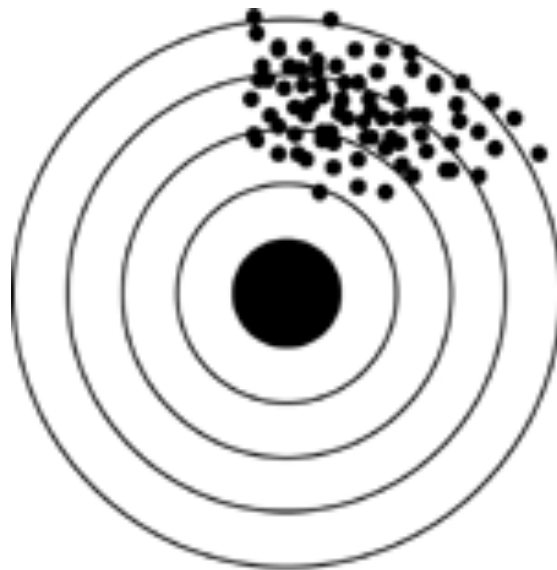
Validity and Reliability

- Reliability

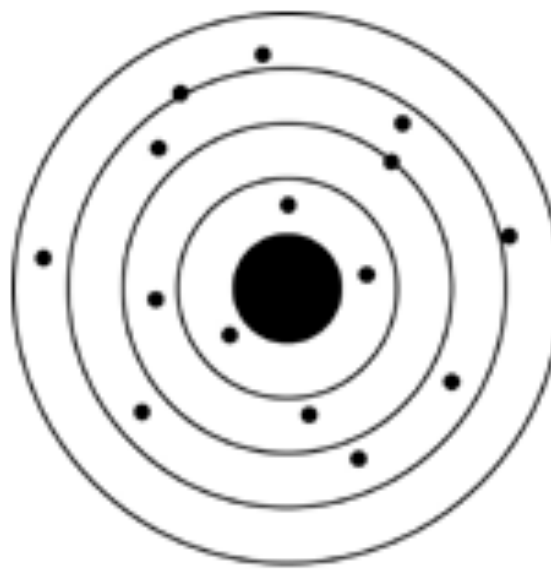
- Does the resulting data yield consistent results?

- Validity

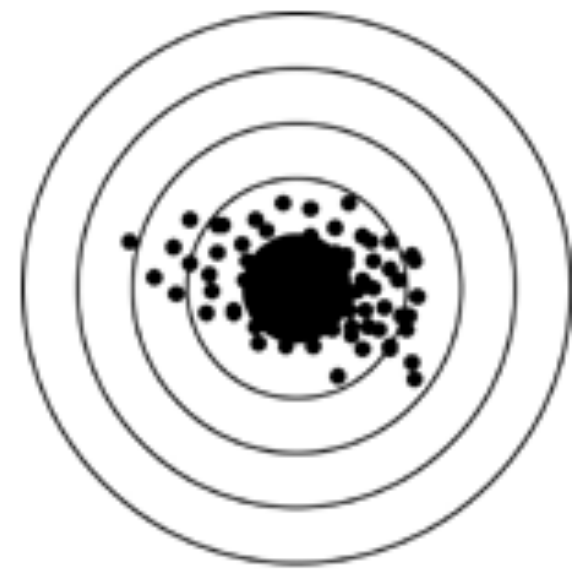
- How useful is this technique in measuring the outcomes?



Reliable but Not Valid

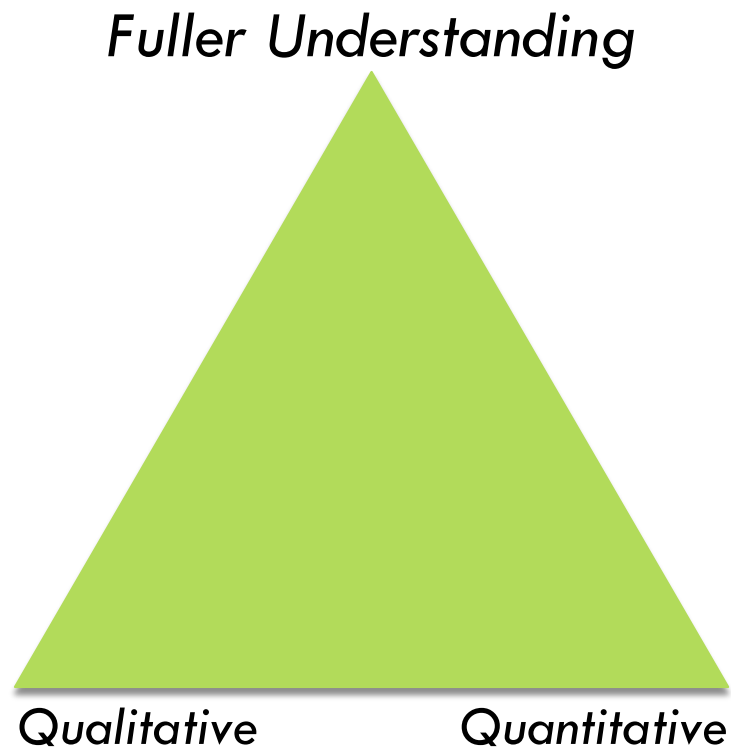


Valid but Not Reliable



Valid and Reliable

Method Triangulation



- Method triangulation allows different methods to be used to collect information about a topic
 - ▣ Provides a fuller picture of the domain
 - ▣ Boosts reliability and validity
 - ▣ Provides different views on the results
 - ▣ Compare, contrast, complement

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